

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
.....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
.....
COMMISSION NATIONALE D'ORGANISATION DE
L'EXAMEN HIGHER NATIONAL DIPLOMA
EXAM (HND)
.....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
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MINISTRY OF HIGHER EDUCATION
.....
NATIONAL COMMISSION FOR THE
ORGANISATION OF HIGHER NATIONAL
DIPLOMA EXAM (HND) EXAM
.....

National Exam of Higher National Diploma 2024 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)
Paper: Information Systems
Duration: 3 hours

Credit: 6

ANSWER ALL QUESTIONS

SECTION A: SYSTEM ARCHITECTURE 40 MARKS

- 1) What is e-waste? (1 mark)
- 2) What is meant by a green data center? (2 marks)
- 3) How do tablets, such as the iPad, enable the paperless office? (3marks)
- 4) What is meant by ubiquitous computing? How about the Internet of Things? (6 marks)
- 5) What is the difference between IaaS, PaaS, and SaaS? (8 marks)
- 6) Define virtualization. How does it relate to the cloud? (6 marks)
- 7) Define *scalable*. Why is this term important to system developers? (5 marks)
- 8) Describe the differences between two-tiered, three-tiered and n-tiered architectures. (9 marks)

SECTION B: OPERATING SYSTEMS (40 MARKS)

I. MCQs (10 Marks)

Each Correct Answer Carries One (01)

1. CPU fetches the instruction from memory according to the value of:
 - A. program counter
 - B. status register
 - C. instruction register
 - D. program status word
2. A memory buffer used to accommodate a speed differential is called:
 - A. stack pointer
 - B. cache
 - C. accumulator
 - D. disk buffer

3. Which one of the following is the address generated by CPU?
 - A. physical address
 - B. absolute address
 - C. logical address
 - D. none of the mentioned
4. Run time mapping from virtual to physical address is done by:
 - A. memory management unit
 - B. CPU
 - C. PCI
 - D. none of the mentioned
5. Memory management technique in which system stores and retrieves data from secondary storage for use in main memory is called:
 - A. fragmentation
 - B. paging
 - C. mapping
 - D. none of the mentioned
6. Effective access time is directly proportional to:
 - A. page-fault rate
 - B. hit ratio
 - C. memory access time
 - D. none of the mentioned
7. In FIFO page replacement algorithm, when a page must be replaced:
 - A. oldest page is chosen
 - B. newest page is chosen
 - C. random page is chosen
 - D. none of the mentioned
8. Which algorithm chooses the page that has not been used for the longest period of time whenever the page required to be replaced?
 - A. first in first out algorithm
 - B. additional reference bit algorithm
 - C. least recently used algorithm
 - D. counting based page replacement algorithm
9. A process is thrashing if:
 - A. it is spending more time paging than executing
 - B. it is spending less time paging than executing
 - C. page fault occurs
 - D. swapping cannot take place
10. Working set model for page replacement is based on the assumption of:
 - A. modularity
 - B. locality
 - C. globalization All of these

II. Structural Questions (30 Marks)

1.
 - a) What are the three main purposes of an operating system? (3 marks)
 - b) We have stressed the need for an operating system to make efficient use of the computing hardware. When is it appropriate for the operating system to forsake this principle and to "waste" resources? Why is such a system not really wasteful? (3 marks)
2. What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment? (2 marks)
3. Keeping in mind the various definitions of operating system, consider whether the operating system should include applications such as Web browsers and mail programs. Argue both that it should and that it should not, and support your answers. (2 marks)
4. What are the major roles played by a systems analyst on a project team? (4 marks)
5. Compare the role of a systems analyst, business analyst, and infrastructure analyst. (3 marks)
6. Describe the major phases in the SDLC? (6 marks)
7. Which phase in the SDLC is most important? Why? (3 marks)
8. Describe the principal steps in the planning phase. What are the major deliverables? (4 marks)

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 MARKS)

1. What is the trade-offs that project managers must manage? (4 marks)
2.
 - a) What is stakeholder analysis?
 - b) Discuss three stakeholders that would be relevant for most projects. (1+2 = 3 marks)
3.
 - a) What are the steps for assessing economic feasibility?
 - b) Describe each step. (1+2 = 3 marks)
4.
 - a) Describe a "risky" project in terms of technical feasibility.
 - b) Describe a project that would not be considered "risky." (1+1 = 2 marks)
5.
 - a) What is the difference between intangible value and tangible value?
 - b) Give three examples of each. (2+2 = 4 marks)
6. Give three examples of business needs for a system. (3 marks)